00-D-1297/3
April 5, 1974

FEDERAL SPECIFICATION SHEET

FEDERAL SPECIFICATION

DRYING AND CONDITIONING TUMBLERS, LAUNDRY, COMMERCIAL:

ONE DOOR TYPE, END LOADING STYLE, REVERSIBLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification 00-D-1297).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for reversible, one door, end loading tumblers.

1.2 Classification. The tumblers shall be of the size and kind of heat as specified (see 6.2).

Size: 200 or 400 (dry-weight pounds)
Kind of heat: steam, gas, butane, propane, oil, or gas/oil.

2. APPLICABLE DOCUMENTS

2.1 Specifications. The following Federal Specification of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein:

00-D-1297 - Drying and Conditioning Tumblers, Laundry, Commercial: General Specification For

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 3510

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3. REQUIREMENTS

3.1 Design and construction. The tumblers shall be in accordance with table I and shall be equipped to tilt the cylinder for loading and unloading. The tumblers shall be capable of receiving and drying and conditioning tightly packed damp laundry and then discharging same without causing "roping" or "tangling". Arrangement shall also be provided for non-circulation of the heated air. The machine shall be designed to use not less than 70 percent outside air and shall be provided with cool-down controls.

TABLE I - Design requirements

Size		Evaporation	Capacity	
Minimum dry weight capacity (pounds)	Type of Heat	Water evaporated per minute, minimum (pounds)	Conditioned dry weight per hour, (pounds)	Drying dry weight per hour, (pounds)
200	steam	5	1800	600
200	gas, butane, oil, propane, gas/oil	10	2400	800
400	steam	5.5	3000	1200
400	gas, butane, oil, propane, gas/oil	15	4800	1600

3.1.1 Housing. The housing containing the cylinder shall tilt a minimum of 15 degrees above and below the horizontal. The housing shall be so constructed to prevent placement of objects under the tilting portion of the machine.

3.1.2 Tilting mechanism. The tilting mechanism shall be motorized gears or hydraulic or air powered pistons. The tilting assembly shall be so constructed and pivoted that the tumbling portion of the operating cycle will be performed with the cylinder in a horizontal position. Automatic operation shall provide for stopping the rotation of cylinder housing assembly at "loading", "operating", and "unloading" positions for proper operation.

3.1.3 Door. The door shall be not less than 48 inches in diameter. The door shall be a semi or automatically operating type, mounted at the front of the unit. The door shall either move on trolleys in a track or be of the overhead sliding or folding type. The door shall be interlocked with the tilting controls to prevent opening until the cylinder is in the "load" or "unload" position.

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3.1.4 Cylinder. The cylinder shall have a minimum diameter of 66 inches. The cylinder shall be of open-end construction without a door and shall consist of a circular back sheet, a front sheet and perforated or woven side sheets. The cylinder shall be suitably reinforced with not less than six lifting ribs and a frame and shall be supported by rollers equipped with polyurethane treads. The circular back sheet of the cylinder when required shall be securely fastened to a substantial ferrous metal transmitter and trunnions. The trunnions shall rotate in suitable ball or roller bearings. An alternative fabricating method for the cylinder is to use galvanized steel woven wire mesh sheets for the sides in lieu of perforated sheets.

3.1.5 Moisture retention control. Unless otherwise specified (see 6.2), the tumblers shall be provided with a moisture retention control. The control shall be adjustable from 15 to 45 percent moisture retention. When the preset moisture content has been reached, the control shall automatically shut off the heat.

3.1.6 Lint collector. The lint collector shall be a separate corrosion resistant component capable of filtering tumbler exhaust prior to its release to outside atmosphere. The lint collector shall be compatible with the exhaust requirements of the tumbler and shall be connected to the tumbler by adequately sized ductwork. Pressure drop through the lint collector shall not exceed 1/2 inch of water pressure. The lint collector shall not support combustion and shall be either a wet or dry type as specified (see 6.2). Dry type collectors shall be equipped with a fire extinguishing system. Exterior surface temperatures of the lint collector shall conform to the requirements for the exterior surface temperature of the tumbler when the collector is installed in the interior of any building. The intended location of the lint collector shall be as specified (see 6.2).

3.1.7 Surface temperature. During maximum temperature performance, 85 percent of the tumbler's exterior surface shall have temperatures not exceeding 115 F. and 15 percent shall not exceed 200 F. The ambient temperature shall be 80 F. for testing.

3.1.8 Burner. The burner shall be equipped with safety devices as specified (see 3.6.1.28 of 00-D-1297) and shall be provided with controls to maintain an adjustable preset stack operating temperature variable to 225 F. maximum. The machine shall be equipped with controls to provide an optional maximum inlet temperature of 350 F. for processing durable press materials and a dial temperature gauge to indicate inlet temperature. When specified (see 6.2), the tumbler shall be equipped with a manual transfer facility to allow for the use of butane, propane, gas, or oil as the heat media in the event of discontinuance of the normal gas supply. An exhaust system shall be furnished in accordance with the manufacturer's current practice. The exhaust system furnished shall be compatible with the style machine specified.

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3.1.9 Insulation. Insulation used to comply with the exterior surface temperature requirement shall be fire-resistant and shall be resistant to and installed to prevent settling, falling off or separating from the open shell. Metal panels shall be used to enclose insulation attached to the machine exterior, except the doors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Visual examination. In addition to the general classification of defects, the following shall apply:

Examine	Defect	Classification Major	Minor
Construction and workmanship, specific	Housing construction to prevent placement of objects under tilting portion	X	
Electrical assembly	Moisture retention control, missing	X	

4.2 Test items, functional testing. During functional testing, the following items shall be checked for compliance with the specified requirements:

(a) Automatic "load", "operate", and "unload" tilting cycles.
(b) Door and tilt interlock to prevent automatic opening of door until the cylinder is in the load or unload position.
(c) Operation of tumbler allowed only in horizontal cylinder position.
(d) Surface temperatures.

5. PREPARATION FOR DELIVERY (See 00-D-1297)

6. NOTES

6.1 General notes. In addition to the notes contained in the general specification, the following ordering data peculiar to these tumblers are specified.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) The size (see 1.2).
(b) The kind of heat (see 1.2).
(c) When a moisture retention control is not required (see 3.1.5).

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